



Name: _____

Date: _____

Score: / 30

Learning Objectives

- Identify and extend number patterns
- Evaluate algebraic expressions by substitution
- Solve one- and two-step equations
- Represent relationships with tables, graphs, and equations

Simplify each expression completely. Show all steps and circle your final answer.

Number patterns & sequences

1. An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 2$. Find the 9th term.

$$a_1 = 4, d = 2, n = 9$$

Answer: _____

2. Find the common difference d of the arithmetic sequence: 2, 9, 16, ...

$$2, 9, 16, \dots$$

Answer: _____

3. An arithmetic sequence has first term $a_1 = 20$ and common difference $d = 8$. Find the 8th term.

$$a_1 = 20, d = 8, n = 8$$

Answer: _____

4. Find the common difference d of the arithmetic sequence: 4, 14, 24, ...

$$4, 14, 24, \dots$$

Answer: _____

5. An arithmetic sequence has first term $a_1 = 16$ and common difference $d = 2$. Find the 15th term.

$$a_1 = 16, d = 2, n = 15$$

Answer: _____

6. Find the common difference d of the arithmetic sequence: 8, 15, 22, ...

$$8, 15, 22, \dots$$

Answer: _____



Solving one-step equations

7. Solve for x: $5x = 8$

$$5x = 8$$

Answer: _____

8. Solve for x: $x + 6 = 18$

$$x + 6 = 18$$

Answer: _____

9. After earning \$5 more, Maria has \$35 in all. Write and solve an equation to find how much she started with.

$$x + 5 = 35$$

Answer: _____

10. Solve for x: $2x = 5$

$$2x = 5$$

Answer: _____

11. Solve for x: $x + 15 = 22$

$$x + 15 = 22$$

Answer: _____

12. After earning \$4 more, Maria has \$33 in all. Write and solve an equation to find how much she started with.

$$x + 4 = 33$$

Answer: _____

13. Solve for x: $2x = 6$

$$2x = 6$$

Answer: _____

14. Solve for x: $x + 3 = 8$

$$x + 3 = 8$$

Answer: _____



15. After earning \$17 more, Maria has \$42 in all. Write and solve an equation to find how much she started with.

$$x + 17 = 42$$

Answer: _____

Two-step equations

16. Solve for x: $3x + -6 = 3$.

$$3x + -6 = 3$$

Answer: _____

17. A plumber charges \$37 per hour plus a flat \$40 fee. If the total bill is \$133, how many hours did the job take? Solve $37x + 40 = 133$.

$$37x + 40 = 133$$

Answer: _____

18. Solve for x: $2x + 1 = 8$. Check your answer.

$$2x + 1 = 8$$

Answer: _____

19. Solve for x: $5x + -2 = -4$.

$$5x + -2 = -4$$

Answer: _____

20. A plumber charges \$27 per hour plus a flat \$25 fee. If the total bill is \$266, how many hours did the job take? Solve $27x + 25 = 266$.

$$27x + 25 = 266$$

Answer: _____

21. Solve for x: $4x + -1 = 13$. Check your answer.

$$4x + -1 = 13$$

Answer: _____

22. Solve for x: $6x + -7 = -7$.

$$6x + -7 = -7$$

Answer: _____



23. A plumber charges \$49 per hour plus a flat \$27 fee. If the total bill is \$175, how many hours did the job take? Solve $49x + 27 = 175$.

$$49x + 27 = 175$$

Answer: _____

24. Solve for x: $4x + 6 = 14$. Check your answer.

$$4x + 6 = 14$$

Answer: _____

Evaluating expressions

25. Evaluate $8x + 2$ when $x = 6$.

$$8x + 2, x = 6$$

Answer: _____

26. A plumber charges \$35 per hour plus a \$74 service fee. What is the total cost for 5 hours? Evaluate $35x + 74$ at $x = 5$.

$$35x + 74, x = 5$$

Answer: _____

27. Evaluate $3x^2 + -4$ when $x = 5$.

$$3x^2 + -4, x = 5$$

Answer: _____

28. Evaluate $8x + -1$ when $x = 4$.

$$8x - 1, x = 4$$

Answer: _____

29. A plumber charges \$56 per hour plus a \$32 service fee. What is the total cost for 1 hours? Evaluate $56x + 32$ at $x = 1$.

$$56x + 32, x = 1$$

Answer: _____

30. Evaluate $4x^2 + 9$ when $x = 3$.

$$4x^2 + 9, x = 3$$

Answer: _____





Topics: Number patterns & sequences, Two-step equations, Solving one-step equations, Evaluating expressions. All answers verified by independent computation.

Solutions



Number patterns & sequences

1. An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 2$. Find the 9th term.

$$a_1 = 4, d = 2, n = 9$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_9 = 4 + (9 - 1)(2).$$

$$\rightarrow = 4 + 8 * 2 = 4 + 16 = 20.$$

Answer: $a_9 = 20$

2. Find the common difference d of the arithmetic sequence: 2, 9, 16, ...

$$2, 9, 16, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 9 - 2 = 7$.

→ Verify: $9 - 2 = 16 - 9 = 7$. The sequence is arithmetic.

Answer: $d = 7$

3. An arithmetic sequence has first term $a_1 = 20$ and common difference $d = 8$. Find the 8th term.

$$a_1 = 20, d = 8, n = 8$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_8 = 20 + (8 - 1)(8).$$

$$\rightarrow = 20 + 7 * 8 = 20 + 56 = 76.$$

Answer: $a_8 = 76$

4. Find the common difference d of the arithmetic sequence: 4, 14, 24, ...

$$4, 14, 24, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 14 - 4 = 10$.

→ Verify: $14 - 4 = 24 - 14 = 10$. The sequence is arithmetic.

Answer: $d = 10$

5. An arithmetic sequence has first term $a_1 = 16$ and common difference $d = 2$. Find the 15th term.

$$a_1 = 16, d = 2, n = 15$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_{15} = 16 + (15 - 1)(2).$$

$$\rightarrow = 16 + 14 * 2 = 16 + 28 = 44.$$

Answer: $a_{15} = 44$

6. Find the common difference d of the arithmetic sequence: 8, 15, 22, ...

$$8, 15, 22, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 15 - 8 = 7$.

→ Verify: $15 - 8 = 22 - 15 = 7$. The sequence is arithmetic.

Answer: $d = 7$



Solving one-step equations

7. Solve for x : $5x = 8$

$$5x = 8$$

→ Divide both sides by 5 to isolate x .

$$\rightarrow x = 8 / 5 = 8/5.$$

Answer: $x = 8 \div 5 = 8/5$

8. Solve for x : $x + 6 = 18$

$$x + 6 = 18$$

→ Subtract 6 from both sides to isolate x .

$$\rightarrow x = 18 - 6 = 12.$$

→ Check: $12 + 6 = 18$. True.

Answer: $x = 18 - (6) = 12$

9. After earning \$5 more, Maria has \$35 in all. Write and solve an equation to find how much she started with.

$$x + 5 = 35$$

→ Let x = starting amount. Equation: $x + 5 = 35$.

→ Subtract 5 from both sides: $x = 35 - 5 = 30$.

→ Maria started with \$30.

Answer: $x = 35 - (5) = 30$

10. Solve for x : $2x = 5$

$$2x = 5$$

→ Divide both sides by 2 to isolate x .

$$\rightarrow x = 5 / 2 = 5/2.$$

Answer: $x = 5 \div 2 = 5/2$

11. Solve for x : $x + 15 = 22$

$$x + 15 = 22$$

→ Subtract 15 from both sides to isolate x .

$$\rightarrow x = 22 - 15 = 7.$$

→ Check: $7 + 15 = 22$. True.

Answer: $x = 22 - (15) = 7$

12. After earning \$4 more, Maria has \$33 in all. Write and solve an equation to find how much she started with.

$$x + 4 = 33$$

→ Let x = starting amount. Equation: $x + 4 = 33$.

→ Subtract 4 from both sides: $x = 33 - 4 = 29$.

→ Maria started with \$29.

Answer: $x = 33 - (4) = 29$



13. Solve for x : $2x = 6$

$$2x = 6$$

→ Divide both sides by 2 to isolate x .

$$\rightarrow x = 6 / 2 = 3.$$

Answer: $x = 6 \div 2 = 3$

14. Solve for x : $x + 3 = 8$

$$x + 3 = 8$$

→ Subtract 3 from both sides to isolate x .

$$\rightarrow x = 8 - 3 = 5.$$

→ Check: $5 + 3 = 8$. True.

Answer: $x = 8 - (3) = 5$

15. After earning \$17 more, Maria has \$42 in all. Write and solve an equation to find how much she started with.

$$x + 17 = 42$$

→ Let x = starting amount. Equation: $x + 17 = 42$.

→ Subtract 17 from both sides: $x = 42 - 17 = 25$.

→ Maria started with \$25.

Answer: $x = 42 - (17) = 25$



Two-step equations

16. Solve for x : $3x + -6 = 3$.

$$3x + -6 = 3$$

→ Subtract -6 from both sides: $3x = 9$.

→ Divide by 3: $x = 3$.

Answer: $x = 9 \div 3 = 3$

17. A plumber charges \$37 per hour plus a flat \$40 fee. If the total bill is \$133, how many hours did the job take?
Solve $37x + 40 = 133$.

$$37x + 40 = 133$$

→ Subtract \$40: $37x = 93$. Divide by 37: $x = 93/37$ hours.

Answer: $x = 93 \div 37 = 93/37$

18. Solve for x : $2x + 1 = 8$. Check your answer.

$$2x + 1 = 8$$

→ Subtract 1 from both sides: $2x = 7$.

→ Divide both sides by 2: $x = 7/2$.

Answer: $x = 7 \div 2 = 7/2$

19. Solve for x : $5x + -2 = -4$.

$$5x + -2 = -4$$

→ Subtract -2 from both sides: $5x = -2$.

→ Divide by 5: $x = -2/5$.

Answer: $x = -2 \div 5 = -2/5$

20. A plumber charges \$27 per hour plus a flat \$25 fee. If the total bill is \$266, how many hours did the job take?
Solve $27x + 25 = 266$.

$$27x + 25 = 266$$

→ Subtract \$25: $27x = 241$. Divide by 27: $x = 241/27$ hours.

Answer: $x = 241 \div 27 = 241/27$

21. Solve for x : $4x + -1 = 13$. Check your answer.

$$4x + -1 = 13$$

→ Subtract -1 from both sides: $4x = 14$.

→ Divide both sides by 4: $x = 7/2$.

Answer: $x = 14 \div 4 = 7/2$



22. Solve for x: $6x + -7 = -7$.

$$6x + -7 = -7$$

→ Subtract -7 from both sides: $6x = 0$.

→ Divide by 6: $x = 0$.

Answer: $x = 0 \div 6 = 0$

23. A plumber charges \$49 per hour plus a flat \$27 fee. If the total bill is \$175, how many hours did the job take?
Solve $49x + 27 = 175$.

$$49x + 27 = 175$$

→ Subtract \$27: $49x = 148$. Divide by 49: $x = 148/49$ hours.

Answer: $x = 148 \div 49 = 148/49$

24. Solve for x: $4x + 6 = 14$. Check your answer.

$$4x + 6 = 14$$

→ Subtract 6 from both sides: $4x = 8$.

→ Divide both sides by 4: $x = 2$.

Answer: $x = 8 \div 4 = 2$



Evaluating expressions

25. Evaluate $8x + 2$ when $x = 6$.

$$8x + 2, x = 6$$

→ Substitute $x = 6$: $8(6) + 2 = 48 + 2 = 50$.

Answer: $8(6) + 2 = 48 + 2 = 50$

26. A plumber charges \$35 per hour plus a \$74 service fee. What is the total cost for 5 hours? Evaluate $35x + 74$ at $x = 5$.

$$35x + 74, x = 5$$

→ Total = $35(5) + 74 = 175 + 74 = \249 .

Answer: $35(5) + 74 = 175 + 74 = 249$

27. Evaluate $3x^2 + -4$ when $x = 5$.

$$3x^2 + -4, x = 5$$

→ Substitute $x = 5$: $3(5)^2 + -4 = 3(25) + -4 = 75 + -4 = 71$.

Answer: $3(5)^2 + -4 = 75 + -4 = 71$

28. Evaluate $8x + -1$ when $x = 4$.

$$8x - 1, x = 4$$

→ Substitute $x = 4$: $8(4) + -1 = 32 + -1 = 31$.

Answer: $8(4) + -1 = 32 + -1 = 31$

29. A plumber charges \$56 per hour plus a \$32 service fee. What is the total cost for 1 hours? Evaluate $56x + 32$ at $x = 1$.

$$56x + 32, x = 1$$

→ Total = $56(1) + 32 = 56 + 32 = \$88$.

Answer: $56(1) + 32 = 56 + 32 = 88$

30. Evaluate $4x^2 + 9$ when $x = 3$.

$$4x^2 + 9, x = 3$$

→ Substitute $x = 3$: $4(3)^2 + 9 = 4(9) + 9 = 36 + 9 = 45$.

Answer: $4(3)^2 + 9 = 36 + 9 = 45$

